

(iii) Use your answer in (b)(ii) to determine the amount n of gas in the sample.

$$n = \dots\dots\dots \text{mol} \quad [1]$$

- (c) The root-mean-square (r.m.s.) speed of the molecules of the gas is 1900 ms^{-1} when pV is equal to 270 J .

Determine the mass, in u , of one molecule of the gas, where u is the unified atomic mass unit.

$$\text{mass} = \dots\dots\dots u \quad [4]$$

[Total: 10]

